

AGGREGATE DEMAND

Understanding the Four Key Components

$$AD = C + I + G + (X - M)$$

What is Aggregate Demand?

Aggregate Demand (AD) is the total demand for all goods and services in an economy at a given price level and time period. It represents the total spending by all sectors of the economy.

$$AD = C + I + G + (X - M)$$

C

Consumption

I

Investment

G

Government Spending

X-M

Net Exports

The Aggregate Demand Curve



Why does the AD curve slope downward?

Wealth Effect

Higher prices reduce real wealth, lowering consumer spending

Interest Rate Effect

Higher prices $\rightarrow$ higher interest rates $\rightarrow$ lower investment

Exchange Rate Effect

Higher prices $\rightarrow$ stronger currency $\rightarrow$ fewer exports

C — Consumption

Largest Component (~60–70% of GDP)



Consumption refers to total spending by households on goods and services — both durable (cars, appliances) and non-durable (food, clothing) goods as well as services.

Key Determinants of Consumption

Disposable Income

Higher income → more spending

Consumer Confidence

Optimism boosts consumption

Interest Rates

Lower rates → cheaper borrowing

Wealth Effect

Rising asset prices increase spending

Inflation Expectations

Expected inflation may accelerate buying

Tax Policy

Tax cuts raise disposable income

I — Investment

Business Spending on Capital (~15–20% of GDP)



Investment is spending by businesses on physical capital — machinery, equipment, buildings, and inventory — plus residential construction. It is the most volatile component of AD.

Types of Investment

1. Business Fixed Investment
(machinery, equipment, software)

2. Residential Investment
(housing construction)

3. Inventory Investment
(change in stock levels)

Key Determinants

Interest Rates

Lower rates reduce cost of borrowing

Business Confidence

Optimism about future profits

Technology & Innovation

New tech spurs capital spending

Corporate Tax Rates

Lower taxes boost retained earnings

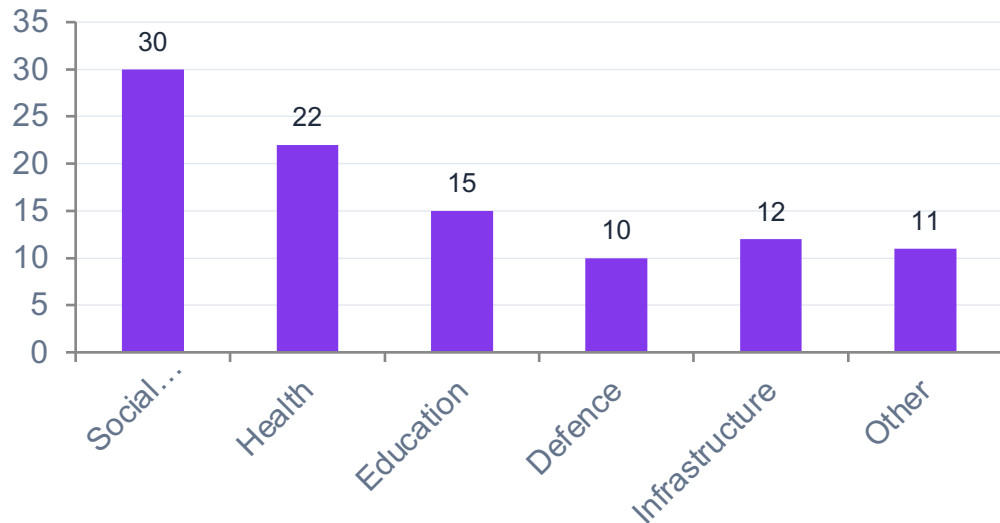
G — Government Spending

Fiscal Policy Tool (~20–25% of GDP)



Government Spending includes all public sector purchases of goods and services — infrastructure, defence, education, healthcare. Note: Transfer payments (pensions, subsidies) are NOT included as they don't involve direct production.

Illustrative Composition of Government Spending (%)



Fiscal Policy Approaches

Expansionary Fiscal Policy

Increase G to boost AD during recession

Contractionary Fiscal Policy

Decrease G to reduce AD during inflation

Multiplier Effect

Each £/\$ of G spending multiplies through economy

$(X - M)$ — Net Exports

Exports minus Imports • Links AD to the Global Economy



Exports (X)

- Goods and services sold abroad
- Adds to domestic AD
- Boosted by weaker domestic currency
- Affected by foreign income levels

Imports (M)

- Goods and services bought from abroad
- Subtracted from domestic AD
- Rise with domestic income and consumer demand
- Boosted by stronger domestic currency

Key Determinants of Net Exports

Exchange Rates

Depreciation → cheaper exports,
pricier imports → NX rises

Relative Inflation

Lower domestic inflation → more
competitive exports

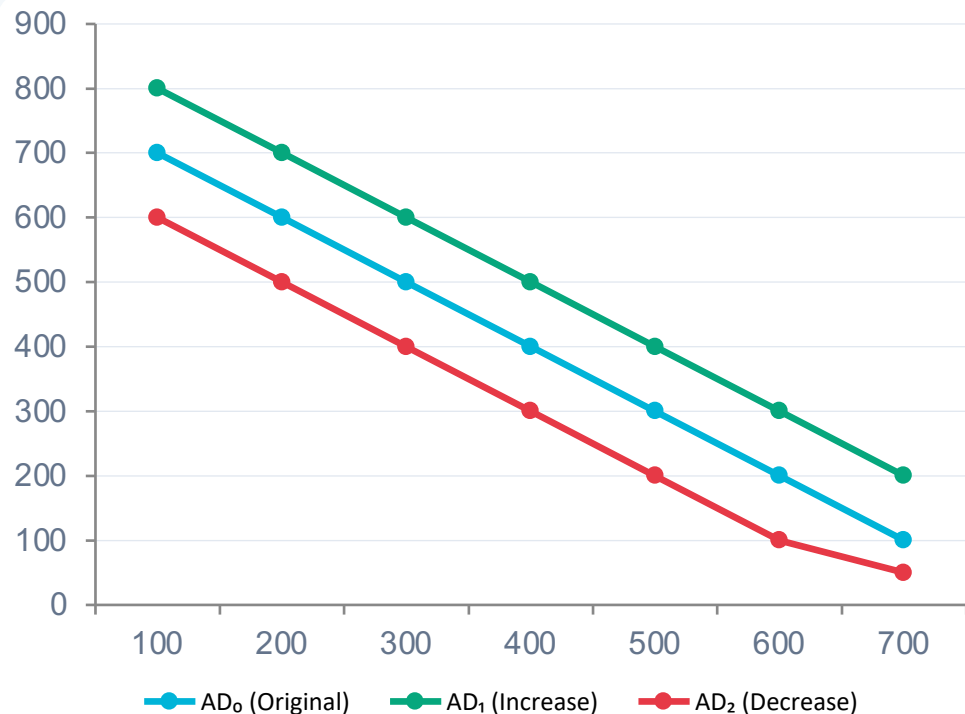
Foreign GDP Growth

Richer trading partners buy more
of our exports

Trade Policies

Tariffs, quotas, and trade
agreements shape NX

Shifts in the AD Curve



→ AD Shifts RIGHT (Increase)

- ↑ Consumer confidence
- ↑ Government spending
- ↑ Money supply / ↓ interest rates
- ↑ Foreign income (boosts exports)
- ↑ Investment expectations

← AD Shifts LEFT (Decrease)

- ↑ Taxes reducing disposable income
- ↑ Interest rates
- ↓ Business confidence
- Currency appreciation (hurts exports)
- ↓ Wealth / falling asset prices

Summary: Components of AD

Component	Formula	Main Determinants	% of GDP (approx)
Consumption (C)	$C = f(Y_d, r, W)$	Income, interest rates, wealth, confidence	60–70%
Investment (I)	$I = f(r, \text{profit expectations})$	Interest rates, business confidence, technology	15–20%
Government Spending (G)	G = set by fiscal policy	Political decisions, economic cycle, public needs	20–25%
Net Exports (X–M)	$NX = X - M$	Exchange rates, foreign income, relative prices	Varies
AD = C + I + G + (X – M) The total demand for output in an economy at all price levels			

Key Takeaways

- 1 Aggregate Demand = $C + I + G + (X - M)$
- 2 Consumption is the largest component, driven by income and confidence
- 3 Investment is the most volatile — highly sensitive to interest rates and expectations
- 4 Government spending is the key lever of fiscal policy to manage the economy
- 5 Net Exports link domestic AD to the global economy via trade and exchange rates
- 6 AD shifts right when spending rises; left when spending falls — affecting output and prices

$$AD = C + I + G + (X - M)$$

The Keynesian Approach to Aggregate Demand

Keynes argued that total spending (Aggregate Demand) determines output and employment in the short run. In a recession, markets do NOT self-correct quickly — government intervention is needed to stimulate AD.

Effective Demand

- Output determined by spending, not production capacity
- Economy can settle below full employment equilibrium
- Demand creates its own supply (contra Say's Law)

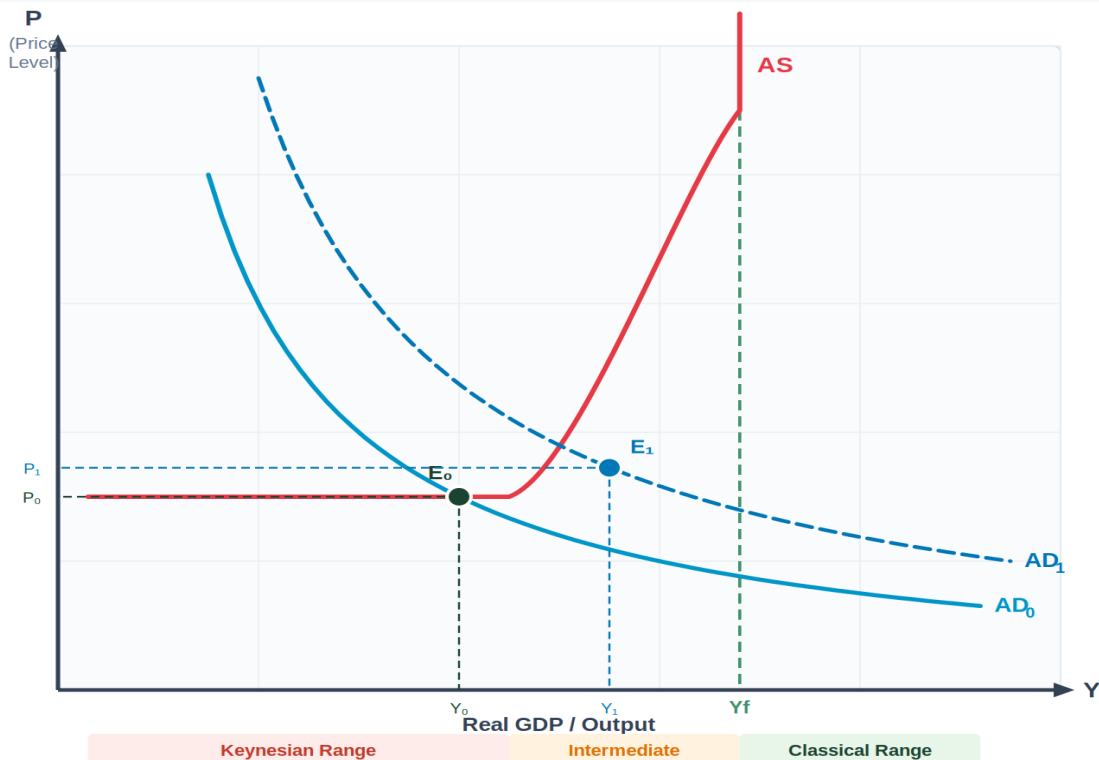
Sticky Wages & Prices

- Wages and prices don't fall quickly in recessions
- Unemployment persists without active intervention
- Markets fail to auto-correct in the short run

Multiplier Effect

- $\Delta Y = \text{Multiplier} \times \Delta G$
- Each unit of govt spending multiplies through economy
- Higher MPC leads to a larger multiplier effect

Keynesian AD–AS Model: The Graph



Keynesian Range (Horizontal AS)

Economy in deep recession with large spare capacity. Output rises without any price increases. Fiscal stimulus is most effective here.

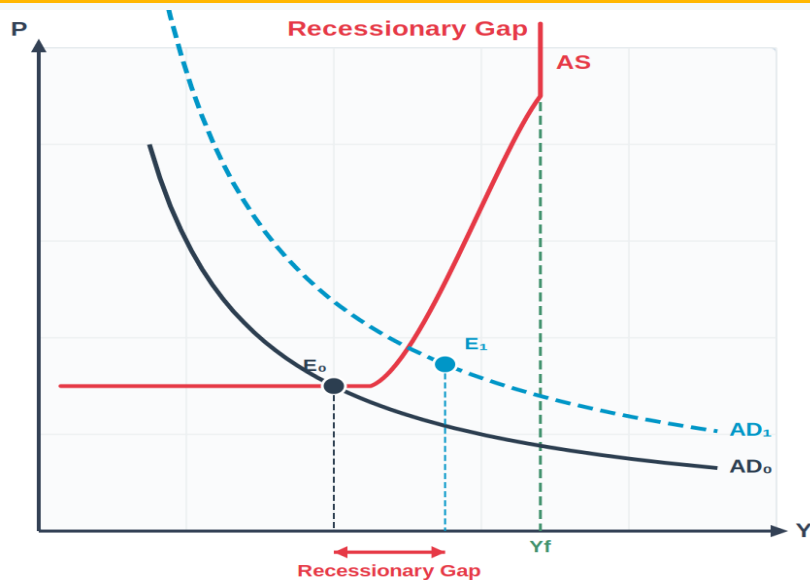
Intermediate Range (Upward-Sloping AS)

Economy approaching full capacity. Both output and price level rise together. Mix of real output gain and inflation occurs.

Classical Range (Vertical AS)

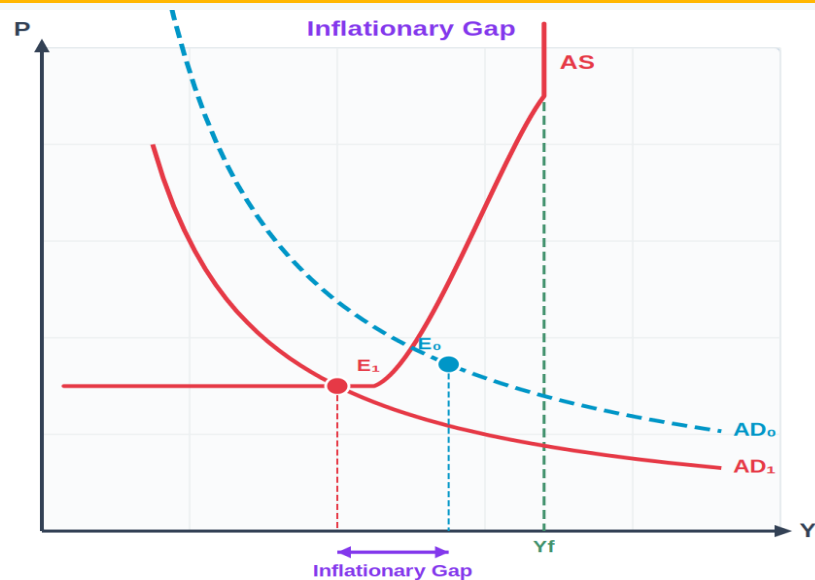
Economy at full employment (Y_f). Any AD increase only raises prices with no real output gain. Classical view prevails here.

AD-AS Equilibrium & Keynesian Policy Implications



Recessionary Gap: Expansionary Policy

- $\uparrow$ Government Spending (G)
- $\downarrow$ Taxes $\rightarrow$ disposable income up ($C \uparrow$)
- $\downarrow$ Interest rates $\rightarrow$ Investment up ($I \uparrow$)
- AD shifts RIGHT $\rightarrow$ closes output gap



Inflationary Gap: Contractionary Policy

- $\downarrow$ Government Spending (G)
- $\uparrow$ Taxes $\rightarrow$ consumption down ($C \downarrow$)
- $\uparrow$ Interest rates $\rightarrow$ Investment down ($I \downarrow$)
- AD shifts LEFT $\rightarrow$ reduces excess demand